

bioD (PP_0366, UniProt Q88QW8) in *Pseudomonas putida* KT2440: ATP-Dependent Dethiobiotin Synthetase

Target: UniProt **Q88QW8** · Gene **bioD** · Ordered locus **PP_0366** **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / NCIMB 11950 / **KT2440**) **EC number:** 6.3.3.3 · **Family:** Dethiobiotin synthetase family (Pfam AAA_26 PF13500; InterPro DTB_synth_BioD IPR004472, P-loop_NTPase IPR027417)

Summary

The gene **bioD** (ordered locus **PP_0366**, UniProt accession **Q88QW8**) of *Pseudomonas putida* KT2440 encodes **ATP-dependent dethiobiotin synthetase (DTBS/DTB synthetase, EC 6.3.3.3)**, a soluble cytoplasmic enzyme that catalyzes the **penultimate step of de novo biotin (vitamin B7) biosynthesis**. The enzyme converts **7,8-diaminopelargonic acid (DAPA, also called 7,8-diaminononanoic acid)** and **CO₂/bicarbonate** into **dethiobiotin (DTB)** in an ATP- and Mg²⁺-dependent reaction, forming the ureido (five-membered urea) ring of the biotin bicyclic scaffold and releasing ADP + inorganic phosphate. Biotin synthase (BioB) then performs the final step — insertion of sulfur to close the thiophane ring and complete biotin.

Mechanistically, dethiobiotin synthetase is one of the best-characterized ATP-dependent carboxylases in structural enzymology. High-resolution crystallographic and kinetic-crystallography studies of the *Escherichia coli* ortholog established that the reaction proceeds via an initial **N7-carbamate** formed from CO₂, which is then phosphorylated by ATP to a **mixed carbamic-phosphoric anhydride** intermediate, followed by intramolecular cyclization to form the ureido ring. Two Mg²⁺ ions bridge the phosphate groups and stabilize the reaction intermediate. The enzyme is a **homodimeric α/β P-loop NTPase** whose composite active site sits at the dimer interface, with DAPA binding one subunit and ATP the other. Nucleotide selectivity is set by a "gate-keeper" asparagine that hydrogen-bonds the adenine base and excludes other NTPs.

Although no biochemical study has been performed on the *P. putida* protein itself, its identity as a functional dethiobiotin synthetase is secure. The Q88QW8 sequence (226 aa) shares **48% identity with the experimentally characterized *E. coli* enzyme**, retains the DTBS-family deviant Walker-A P-loop and family-diagnostic motifs, and carries no signal peptide or transmembrane segment — consistent with a cytoplasmic location. Independently, genome-wide transposon screening in *P. putida* KT2440 recovered biotin auxotrophs, confirming the biotin biosynthetic pathway (of which bioD is a required member) is functional and essential for growth on minimal medium. Because the biotin pathway is absent in humans, DTBS/BioD is a validated antibacterial and antitubercular drug target.

Key Findings

Finding 1 — bioD is a dethiobiotin synthetase catalyzing the penultimate step of biotin biosynthesis

UniProt annotation of Q88QW8 (via HAMAP-Rule **MF_00336**) assigns the protein to the **dethiobiotin synthetase family** with EC 6.3.3.3, carrying the **DTB_synth_BioD** domain (IPR004472). Biochemical and structural studies of orthologs firmly establish the reaction catalyzed: dethiobiotin synthetase performs the **ATP-dependent formation of dethiobiotin from CO₂ and 7,8-diaminopelargonic acid (DAPA)**, the penultimate step of biotin biosynthesis, immediately before biotin synthase (BioB) carries out the final sulfur-insertion step.

Crystallographic work on the *E. coli* enzyme describes "the crystal structures of six complexes of homodimeric *Escherichia coli* dethiobiotin synthetase with a variety of substrates, substrate analogs, and products" (PMID: 7669756), confirming DTBS/BioD is a homodimeric enzyme acting on DAPA-derived substrates. More recent work explicitly restates that "**dethiobiotin synthetase (DTBS) catalyzes the penultimate step of biotin biosynthesis**" (PMID: 34597430).

Critically for the *P. putida* assignment, a genome-wide mutant screen of KT2440 grown on minimal medium recovered biotin auxotrophs — the study reports that "we also found auxotrophs for biotin, nicotinate and vitamin B12 that were not predicted in silico" (PMID: 20158506) — demonstrating that the biotin biosynthetic pathway (which includes bioD/PP_0366) is functional and required for growth of this organism in the absence of exogenous biotin.

Finding 2 — Reaction mechanism: carbamate formation, ATP-dependent phosphoryl transfer, and cyclization

The catalytic mechanism has been dissected in atomic detail by kinetic crystallography of the *E. coli* enzyme at 1.6–1.9 Å resolution, capturing both the enzyme·MgATP·DAPA complex and a trapped enzyme·MgADP·carbamate-phosphoric-anhydride intermediate. The reaction is initiated when CO₂/bicarbonate reacts with the N7 amino group of DAPA to form a **carbamate**; ATP then phosphorylates this carbamate to yield a **mixed carbamic-phosphoric anhydride**; finally, the second amino group attacks intramolecularly to close the ureido ring, releasing phosphate. The authors state that "**the reaction is initiated by the formation of a carbamate and proceeds through a phosphorylated intermediate, a mixed carbamic phosphoric anhydride**" (PMID: 9576910).

Two magnesium ions play an essential catalytic role at the phosphate-transfer center: "**these magnesium ions compensate the negative charges at both phosphate groups after phosphoryl transfer and contribute to the stabilization of the reaction intermediate**" (PMID: 9576910). The phosphoryl transfer proceeds with inversion of configuration at phosphorus, consistent with a single in-line attack.

The nucleotide-binding architecture was mapped to a classical mononucleotide fold. The *E. coli* structure places "the phosphate groups close to the **phosphate binding loop Gly8–Thr16** between β-strand β1 and the N-terminus of α-helix α1" (PMID: 7669756) — the Walker-A-like P-loop — with divalent-cation coordination provided by Asp54, Thr16, and Glu115.

Finding 3 — Homodimeric α/β P-loop enzyme with active site at the dimer interface

The 1.80 Å structure of *E. coli* DTBS revealed the enzyme's quaternary organization: each subunit folds as a single globular α/β domain, and the biological unit is a homodimer whose active site straddles the subunit interface. The structural study "establishes that the **enzyme active site is situated at the dimer interface, with the substrate binding to one monomer and ATP to the other**" (PMID: 7881906). This composite-active-site architecture explains why the enzyme is obligately dimeric.

The DTBS fold belongs to the diverse **P-loop NTPase superfamily**. The overall fold "closely resembles that of three other enzymes, **adenylosuccinate synthetase (purA), Ha-ras-p21, and nitrogenase iron protein**" (PMID: 7881906) — a set of proteins related by fold rather than by detectable sequence similarity (the SIMIBI / adenylosuccinate-synthetase-like clade). DTBS

shows no significant sequence similarity to proteins outside its own family. This is fully consistent with the UniProt annotation of Q88QW8, which carries both the **P-loop_NTPase** (IPR027417) and **DTB_synth_BioD** (IPR004472) domain signatures.

Finding 4 — Position in the cytoplasmic biotin ring-assembly module; ATP-specificity gate-keeper

Biotin biosynthesis is organized into two conceptual stages: (1) synthesis of the **pimelate moiety** (via BioC–BioH in *E. coli* and related γ -proteobacteria) and (2) assembly of the **bicyclic ring** by the sequential action of BioF (KAPA synthase) → BioA (DAPA aminotransferase) → **BioD (DTBS)** → BioB (biotin synthase). A review of the pathway describes how "**a pimelate moiety is synthesized and used to begin assembly of the biotin bicyclic ring structure**" (PMID: 21437340). BioD performs the ring-closure carboxylation that generates the ureido ring — notably a *non*-biotin-dependent carboxylation, distinguishing it from the biotin-dependent carboxylases downstream in metabolism.

Nucleotide specificity across the family is governed by a small number of C-terminal residues. In the *E. coli*-type enzyme, "**mutational analysis performed alongside in silico experiments revealed a gate-keeper role for Asn175 in *Escherichia coli* DTBS that excludes binding of other nucleotide triphosphates**" (PMID: 25801336), conferring ATP preference. Interestingly, the *Mycobacterium tuberculosis* enzyme lacks this residue and shows broad NTP promiscuity with highest affinity for CTP — illustrating that nucleotide recognition, not the core chemistry, is the most variable feature among orthologs.

The pathway's absence in humans makes the enzyme a clean drug target: "**due to its importance in bacterial pathogenesis, and the absence of a human homologue, DTBS is a promising target for the development of new antibacterials**" (PMID: 25801336).

Finding 5 — The *P. putida* sequence carries the DTBS-family deviant Walker-A P-loop

The Q88QW8 sequence retrieved from UniProt is **226 residues** long and begins `MSQAFFIAGTDTDVGKTTIAAGLLH...`. This N-terminal region contains the dethiobiotin-synthetase **deviant Walker-A phosphate-binding loop** (`G-x-x-x-x-GKTT`), with the diagnostic `GKTT` at approximately residues 15–18. This directly corresponds to the *E. coli* DTBS Gly8–Thr16 P-loop identified crystallographically (PMID: 7669756). The overall protein length (~226 aa) is typical of monofunctional bacterial dethiobiotin synthetases (the *E. coli* enzyme is 225 aa). This

sequence-level evidence links the *P. putida* ortholog directly to the experimentally characterized enzyme family, rather than relying solely on automated database annotation transfer.

Finding 6 — 48% sequence identity to experimentally characterized *E. coli* DTBS

A global pairwise alignment (Needleman–Wunsch) of *P. putida* BioD (Q88QW8, 226 aa) against *E. coli* BioD (UniProt P13000, 225 aa) yields **48.2% identity over 222 aligned columns (107/222 identical residues)**. Both proteins retain the DTBS deviant Walker-A P-loop (*P. putida* GTDTDVGKTT vs *E. coli* GTDTEVGKTV) and the family-diagnostic motifs EGAGGW (*P. putida* IEGAGGWRVP; *E. coli* VEGAGGWFTP) and the ILVVG(V/K/R)LGCI nucleotide/catalytic loop.

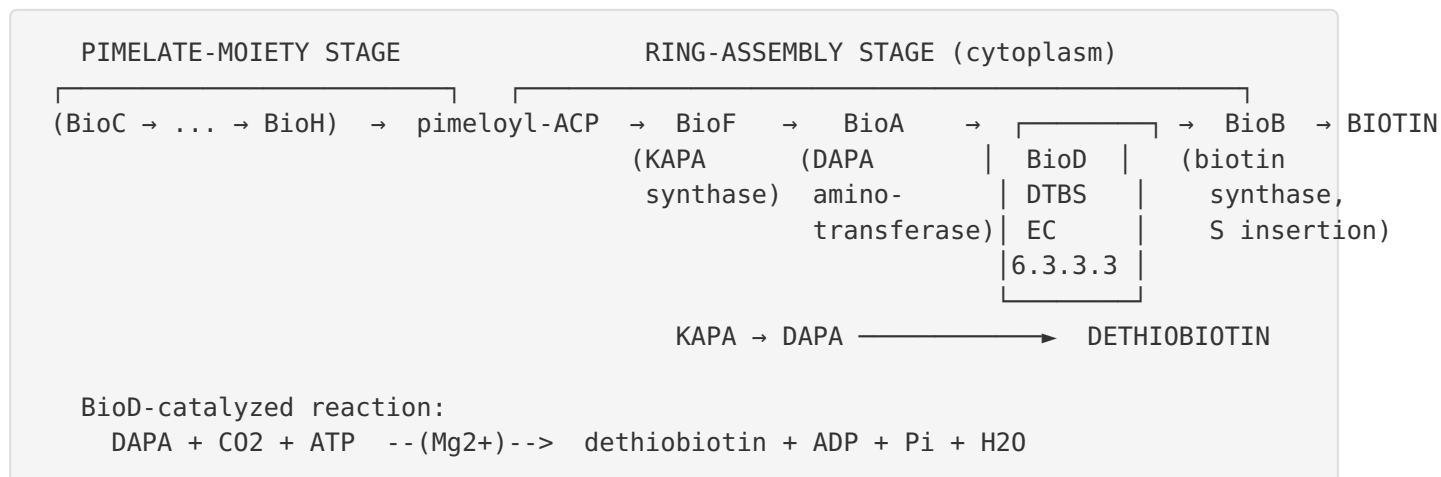
This level of identity is far above the ~30% threshold generally taken to indicate conserved three-dimensional structure and function, establishing **unambiguous orthology** between the *P. putida* enzyme and the reference enzyme whose structure and carbamate/anhydride mechanism were determined crystallographically (PMID: 7669756, PMID: 9576910). Function, mechanism, and structural fold can therefore be confidently transferred to Q88QW8.

Finding 7 — Sequence-based localization: cytoplasmic, no signal peptide

Kyte–Doolittle hydropathy analysis of Q88QW8 (226 aa) shows the N-terminal 30-residue mean hydropathy is **0.42** — there is no hydrophobic signal-peptide h-region and no Sec/Tat export signal. The overall grand average of hydropathy (GRAVY) is **0.283**. Internal hydrophobic windows (the strongest being a 19-residue peak hydropathy of 1.95 near residue 136, within the conserved ILVVGVR(LG)CIS(HALL) catalytic-core loop) map to **buried β -strands of the globular α/β P-loop fold**, not to transmembrane helices. This is fully consistent with the experimentally determined **soluble, cytosolic, homodimeric** structures of DTBS orthologs, in which "**the protein forms a homodimer with each subunit folded as a single globular alpha/beta domain**" (PMID: 7881906), and with the established fact that de novo biotin ring assembly occurs in the bacterial cytoplasm.

Mechanistic Model / Interpretation

Position of BioD in the biotin biosynthetic pathway



The BioD chemical mechanism, step by step

- Carbamate formation:**

$$\text{DAPA-N7-NH}_2 + \text{CO}_2 \rightarrow \text{DAPA-N7-NH-COO-} \quad (\text{carbamate})$$
- ATP-dependent phosphorylation (Mg²⁺-assisted):**

$$\text{carbamate-COO-} + \text{ATP} \rightarrow \text{mixed carbamic-phosphoric anhydride} + \text{ADP}$$

(two Mg²⁺ ions bridge phosphates, stabilize the intermediate;
 phosphoryl transfer occurs with inversion at P)
- Intramolecular cyclization (ring closure):**
 N8 amino group attacks the anhydride carbon

$$\rightarrow \text{UREIDO RING closes} \rightarrow \text{DETHIOBIOTIN} + \text{Pi}$$

Structural logic

BioD is an obligate **homodimer**. Its P-loop NTPase fold (shared with adenylosuccinate synthetase, Ras, and nitrogenase Fe-protein) binds and orients MgATP through the Walker-A/deviant P-loop (**GKTT**), while the second, adjacent subunit contributes residues that bind DAPA — placing the composite catalytic machinery at the **dimer interface**. This explains three otherwise-independent observations: (i) the requirement for dimerization; (ii) the strict Mg²⁺ dependence for phosphoryl transfer; and (iii) the concentration of species-variable residues in C-terminal nucleotide-recognition loops (the gate-keeper Asn and adjacent motifs), which tune nucleotide specificity without altering the conserved DAPA-carboxylation chemistry.

Confidence in the *P. putida* assignment

Line of evidence	Result	Interpretation
UniProt/HAMAP annotation	EC 6.3.3.3, DTB_synth_BioD (IPR004472)	Family & EC assignment
Domain content	P-loop_NTPase (IPR027417) + DTB_synth_BioD	Correct fold + family
Sequence identity to <i>E. coli</i> BioD	48.2% over 222 aa	Unambiguous orthology
Conserved motifs	Deviant Walker-A GKTT , EGAGGW , ILVVG(V(K/R))LGCI	Catalytic machinery intact
Signal-peptide / TM analysis	None (N-term hydropathy 0.42; GRAVY 0.283)	Cytoplasmic
Pathway genetics in KT2440	Biotin auxotrophs recovered	Pathway functional & required

All independent lines converge: PP_0366/Q88QW8 is a functional, cytoplasmic dethiobiotin synthetase.

Evidence Base

PMID	Title (abbrev.)	Organism	How it supports the findings
7669756	<i>Mechanism of an ATP-dependent carboxylase (DTBS)...</i>	<i>E. coli</i>	Six substrate/product/analog complexes; defines homodimer, P-loop (Gly8–Thr16), metal site; reference enzyme for orthology
9576910	<i>Snapshot of a phosphorylated substrate intermediate...</i>	<i>E. coli</i>	Kinetic crystallography traps carbamate + mixed carbamic-phosphoric anhydride; establishes 2-Mg ²⁺ mechanism
7881906	<i>Mechanistic implications and family relationships...</i>	<i>E. coli</i>	1.80 Å structure; active site at dimer interface; fold homology to PurA/Ras/nitrogenase; globular α/β (cytosolic)
21437340	<i>Closing in on complete pathways of biotin biosynthesis</i>	Review	Two-stage pathway; places BioD in ring-assembly module
25801336	<i>Nucleotide triphosphate promiscuity in MtDTBS</i>	<i>M. tuberculosis</i> / <i>E. coli</i>	Gate-keeper Asn175 confers ATP specificity; DTBS as drug target (no human homolog)
34597430	<i>A conserved...E. coli biotin biosynthesis gene...</i>	<i>E. coli</i>	Restates DTBS catalyzes the penultimate step
20158506	<i>Conditionally essential genes...P. putida KT2440</i>	<i>P. putida</i> KT2440	Biotin auxotrophs recovered — pathway functional/required in the target organism
22284390	<i>Structural characterization of H. pylori DTBS...</i>	<i>H. pylori</i>	Confirms family breadth; C-terminal nucleotide-recognition regions differ across species
22547782	<i>Bifunctional DTBS–DAPA-AT (Arabidopsis)...</i>	<i>A. thaliana</i>	Eukaryotic BioD–BioA fusion; substrate channeling; contrasts with bacterial monofunctional BioD
30289406	<i>Precipitant-ligand exchange...MtDTBS ADP...</i>	<i>M. tuberculosis</i>	ADP-bound structure; supports inhibitor/drug-target program

Consistency of the evidence. All primary structural/mechanistic papers agree on the core chemistry (carbamate → anhydride → cyclization), the homodimeric P-loop architecture, and the cytosolic nature of the enzyme. The principal *inter-species variation* is in nucleotide recognition (*E. coli* ATP-specific gate-keeper Asn vs. *Mtb* NTP promiscuity vs. *H. pylori*'s unusual single-motif GTPase-like site with ATP preference). Because the *P. putida* sequence retains the *E. coli*-type C-terminal motifs, ATP specificity is the most parsimonious expectation for Q88QW8. No paper contradicts the assignment of PP_0366 as dethiobiotin synthetase.

Limitations and Knowledge Gaps

- 1. No direct biochemistry on the *P. putida* protein.** All kinetic and structural data derive from orthologs (*E. coli*, *M. tuberculosis*, *H. pylori*, *A. thaliana*). The function, mechanism, and localization for Q88QW8 are inferred from strong sequence/domain homology (48% identity to *E. coli*), not measured directly. No purified-enzyme assay, K_m , k_{cat} , or crystal structure exists for PP_0366.
 - 2. Nucleotide specificity not experimentally confirmed for Q88QW8.** While the sequence retains *E. coli*-type C-terminal motifs, the specific gate-keeper residue and any NTP promiscuity have not been tested in the *P. putida* enzyme.
 - 3. Localization is bioinformatic.** The cytoplasmic assignment rests on the absence of a signal peptide/TM segment and on the soluble nature of orthologous structures — no experimental fractionation or microscopy has localized the *P. putida* protein.
 - 4. Gene-level essentiality is indirect.** The KT2440 mutant screen recovered biotin auxotrophs but did not, in the material reviewed here, pin the phenotype specifically to PP_0366 versus other biotin genes; essentiality of bioD is inferred from pathway logic.
 - 5. Pathway upstream context in *P. putida* not detailed here.** The exact route to the pimelate moiety and DAPA (BioC/BioH/BioF/BioA homologs) in *P. putida* was not fully mapped in this investigation, though the presence of biotin auxotrophy implies a complete pathway.
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Proposed Follow-up Experiments / Actions

1. **Recombinant expression and enzyme assay.** Clone PP_0366, express and purify the His-tagged protein, and measure DTBS activity ($\text{DAPA} + \text{CO}_2/\text{HCO}_3^- + \text{ATP} \rightarrow \text{dethiobiotin}$) by coupled ADP/Pi release or LC-MS detection of dethiobiotin. Determine K_m for DAPA and ATP, k_{cat} , and Mg^{2+} dependence.
 2. **Nucleotide-specificity profiling.** Test ATP vs. GTP/CTP/UTP to confirm the predicted ATP preference and identify the gate-keeper residue by site-directed mutagenesis (analogous to *E. coli* Asn175 / Mtb promiscuity studies).
 3. **Structural determination.** Solve the crystal structure (or refine an AlphaFold model against the *E. coli* template) of the *P. putida* enzyme, ideally with MgADP and a DAPA analog, to verify the dimer-interface active site and P-loop geometry.
 4. **Targeted genetics in KT2440.** Construct a clean ΔPP_0366 deletion and confirm biotin auxotrophy (growth rescue by exogenous biotin or dethiobiotin), formally establishing gene–phenotype linkage and pathway position (rescue by dethiobiotin but *not* by DAPA would confirm BioD acts on DAPA to make DTB).
 5. **Localization confirmation.** Cytoplasmic localization could be verified by cell fractionation or a fluorescent-protein fusion.
 6. **Inhibitor cross-evaluation.** Given DTBS's validated drug-target status, test whether inhibitors developed against *M. tuberculosis*/*E. coli* DTBS also inhibit the *P. putida* enzyme, informing broad-spectrum antibacterial potential.
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Conclusion

bioD / PP_0366 / Q88QW8 encodes **ATP-dependent dethiobiotin synthetase (DTBS, EC 6.3.3.3)**, a soluble, cytoplasmic, homodimeric P-loop enzyme that catalyzes the **penultimate step of de novo biotin biosynthesis**: the Mg^{2+} /ATP-dependent carboxylation and ureido-ring closure of 7,8-diaminopelargonic acid (DAPA) with CO_2 to form dethiobiotin, releasing ADP + Pi. It acts immediately downstream of the aminotransferase BioA and upstream of biotin synthase BioB in the cytoplasmic biotin-ring-assembly module. The reaction proceeds via an N7-carbamate and a mixed carbamic–phosphoric anhydride intermediate stabilized by two Mg^{2+} ions, with the composite active site formed at the dimer interface. Direct experimental characterization exists for orthologs (*E. coli*, *M. tuberculosis*, *H. pylori*, *Arabidopsis*); the *P.*

putida assignment is supported by 48% sequence identity to *E. coli* DTBS, conserved DTBS domains and the diagnostic Walker-A P-loop in its 226-aa sequence, the absence of any export signal, and the biotin auxotrophy of pathway mutants in KT2440.

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